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Coaxial electrospinning of continuous PVDF/Bi_2O_3/CeO_2 nanofibre yarns with lead-equivalent X-ray attenuation for radiation-protective healthcare textiles

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ABSTRACT

Lead-based protective garments weighing 2–7 kg impose a musculoskeletal burden on interventional radiology staff. Currently developed alternative shielding materials focus on enabling lightweight, textile-compatible protection; however, no lead-substitute material has yet been produced in a continuous yarn form. This study investigates electrospun yarn production via coaxial and conjugate-polarity configurations. Membrane casting was first used to establish the maximum filler loading at which PVDF and the bismuth/cerium oxides form a homogeneous, electrospinnable solution; loadings up to 90 wt% were achieved, substantially exceeding the practical limits of single-fluid electrospinning. A DMSO/acetone solvent system replaced toxic DMF while promoting a high β-phase content in the PVDF matrix. In the resulting core-sheath yarns, the core Bi_2O_3/CeO_2 provides shielding capacity while the PVDF sheath maintains mechanical strength. The composite membrane reached 48.7% X-ray attenuation at 80 kV. This nearly matches the commercial 0.2 mm lead film reference (47.8%). Transferred to the yarn format, the highest-loading yarn (90 wt%) reached an effective linear attenuation coefficient of 0.766 mm^{-1}, exceeding that of the membrane and demonstrating that shielding efficiency is retained in a continuous yarn.

1. Introduction

Medical staff in interventional radiology (IR) are exposed to scattered radiation during fluoroscopically guided procedures [27]. They are different from the patients, who receive acute doses during discrete examinations, and the clinical staff accumulates exposure over the course of an entire career. Prolonged low-dose irradiation has been associated with posterior subcapsular cataracts [32], thyroid disorders [30], and elevated incidence of specific malignancy subtypes [28], producing an effective personal protective equipment and occupational health imperative. Although medical staff currently followhealth-tracking protocols, periods during which they do not wear the protective apron, owing to its heavy weight, cannot be ignored. The perspective of investigating personal protective equipment's shielding should be reconsidered.

Current investigation of personal protective equipment relies on a lead composite. They design different garments, worn with an added belt, or separately as the apron and vest. Even if the shielding garments provide effective X-ray attenuation, the garments also have a 2–7 kg musculoskeletal burden [6]. Therefore, these problems produce medical staff's back pain, cervical spondylosis, and operator fatigue [24,6]. This previous evidence consistently identifies garment weight and discomfort as the main risks in the current radiation protection. The equipment designed to safeguard health itself can cause primary drivers of PPE non-compliance [16].

The lead-free X-ray attenuation has been researched on dispersed basic materials with high-Z-elements, such as Bi_2O_3 (Z = 83), CeO_2 (Z = 58), WO_3 (Z = 74) [5,13,33,14,26]. These composites can approach the attenuation of equivalent lead thicknesses [5,13,33]. Similar bismuth-loaded flexible composites have also been explored for gamma-ray shielding [34]. However, most of the contribution on x-ray shielding is membrane formats, such as human-made leather, thereby overlooking the broader possibilities afforded by integration with textile technologies. Indeed, the work has been conducted almost exclusively on flat films, cast membranes, and coated substrates [15]. Such geometries cannot be cut, sewn, draped, or laundered as garment construction demands. Therefore, the x-ray shielding materials have been limited to the point at which material fabrication succeeds and shielding performance

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is maximized, and no further. Obviously, redesigns the way of X-ray shielding via the textile processing technologies.

PVDF has been regarded by several researchers as a lightweight, comfortable base polymer for wearable applications, capable of incorporating multiple functionalities [29]. This interest is further motivated by the intrinsic properties of PVDF itself. The electroactive properties of PVDF, specifically the piezoelectric β-phase, offer a potential route to embedded physiological sensing without additional sensor elements [12,35], a capability positioned here as a future research direction, supported by FTIR-confirmed β-phase retention.

This study addresses the identified gap by developing PVDF/Bi_2O_3/CeO_2 composite nanofibre yarns via coaxial electrospinning onto a custom funnel collector, producing a continuous, weavable radiation-shielding yarn format not previously reported in the literature [1,7]). The yarn format was chosen to enable wearable, multifunctional integration. This research investigates a stable electrospinning process for continuous yarn using a DMSO/acetone solvent system free from DMF [31]. Characterization of the morphological, chemical, thermal, and mechanical properties of the electrospinning yarns; the X-ray attenuation performance is confirmed by XCT; and, as the basic polymer Polyvinylidene fluoride (PVDF), the β-phase is evaluated via FTIR as structural evidence of the material's intrinsic piezoelectric potential for future sensing integration.

In the present article, the success of producing a continuous radiation-shielding nanofibre yarn via electrospinning with PVDF and high z elements. Secondly, a coaxial core-sheath electrospinning yarn that resolves the fundamental mechanical constraint on high filler loading [22,4]. The combination of coaxial needles and dual needles in a spinning setup helps increase the nanoparticle concentrations up to 90 wt%. Furthermore, it confirms high β-phase in the PVDF matrix despite composite loading, establishing the material's piezoelectric potential as a basis for future integrated sensing applications.

2. Materials and methods

2.1. Materials and solution preparation

This research selects PVDF powder (Mw 1,800,000 g/mol, Kynar, Arkema) as the base polymer. Bi_2O_3 nanoparticles (80–100 nm) and CeO_2 nanoparticles (10–30 nm) were used as X-ray attenuating fillers with DMSO and acetone (both analytical grade) as the solvent system in place of the conventional DMF to eliminate reproductive toxicity [31].

The dispersion was conducted via probe ultrasonication. Bismuth oxide and cerium oxide were combined at a 2:1 mass ratio (molar ratio of 1:1.35), informed by Li et al. [20], to achieve a synergistic shielding effect. The PVDF and bismuth oxide/cerium oxide have three levels of ratio: the PVDF:BC mass ratio is 5:5, 3:7, and 1:9 (corresponding to 50, 70, and 90 wt% BC. The solvent used DMSO with acetone in ratio 1:2 [17]. A separate pure PVDF solution was prepared as a coaxial sheath. The complete solution preparation workflow is illustrated in Fig. 1.

2.2. Electrospinning and yarn formation

Electrospinning was conducted at a controlled temperature of 20 ± 2 °C and 40 ± 5% relative humidity. A dual-needle opposed-polarity configuration was employed (±6 kV) [11], with needles 17 cm apart and each positioned 17 cm from the funnel collector, forming an equilateral triangle shown in Fig. 2. This geometry eliminated the requirement for a grounded collector whilst providing a symmetric electric field that draws fibres from both needles towards the funnel simultaneously. Solutions were delivered at 3.5 mL h^{-1} per needle via syringe pump.

The collector comprised a custom aluminium funnel (diameter 77 mm) [19]operated at 200 rpm, with a winding bobbin (diameter 20 mm) at a winding speed of 8 rpm. The tip to distence located the dual needles and funnel collector each other is 17 cm. The continuous fibre membrane is spun on the funnel edge, then wound on a bobbin positioned on the opposite side of the funnel. A continuous draw force is applied to the center of the funnel's membrane, deforming it into a yarn cone, from which twisted yarn is continuously extracted.

Six yarn samples were produced in two electrospinning formats (membrane and yarn), as summarised in Table 1. A single-needle configuration with G22 needle (i.d. 0.41 mm) was employed for the pure PVDF membrane (PVDF-F), composite membrane (PVDFBC-F), and pure PVDF yarn (PVDF-Y), as the unfilled and single-component solutions showed sufficient spinnability at the target flow rate. The yarn samples (PVDFBC-Y55 to PVDFBC-Y19) is a coaxial yarn with high filler loading, and the syringe solution fed through the needles that inner channel (G22, i.d. 0.41 mm) and pure PVDF solution through the outer channel (G17, i.d. 1.07 mm) at 1 mL h^{-1} per channel, forming a protective sheath layer.

2.3. Characterisation methods

2.3.1. Scanning electron microscopy (SEM)

The bead-free fibre formation across all yarn compositions via SEM. The smooth fibre twisted as a yarn shows the successful solvent making, and the DMSO/acetone ratio. This is an appropriate solution viscosity for stable electrospinning system. The effect of increasing nanoparticle loading on fibre surface topology was assessed, with particular attention to nanoparticle dispersion homogeneity and the presence of surface aggregation. Fibre twist morphology within the yarn structure was also examined to verify the effectiveness of the funnel collection mechanism in producing cohesive twisted yarn.

2.3.2. X-ray computed tomography (XCT)

X-ray computed tomography was performed using a ZEISS Xradia

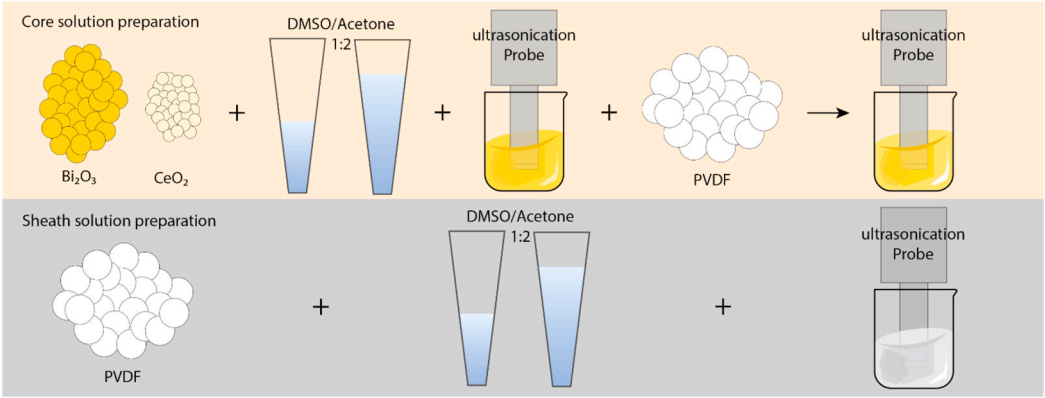


Fig. 1. Schematic of solution preparation for the PVDF/Bi_2O_3/CeO_2 electrospinning system.

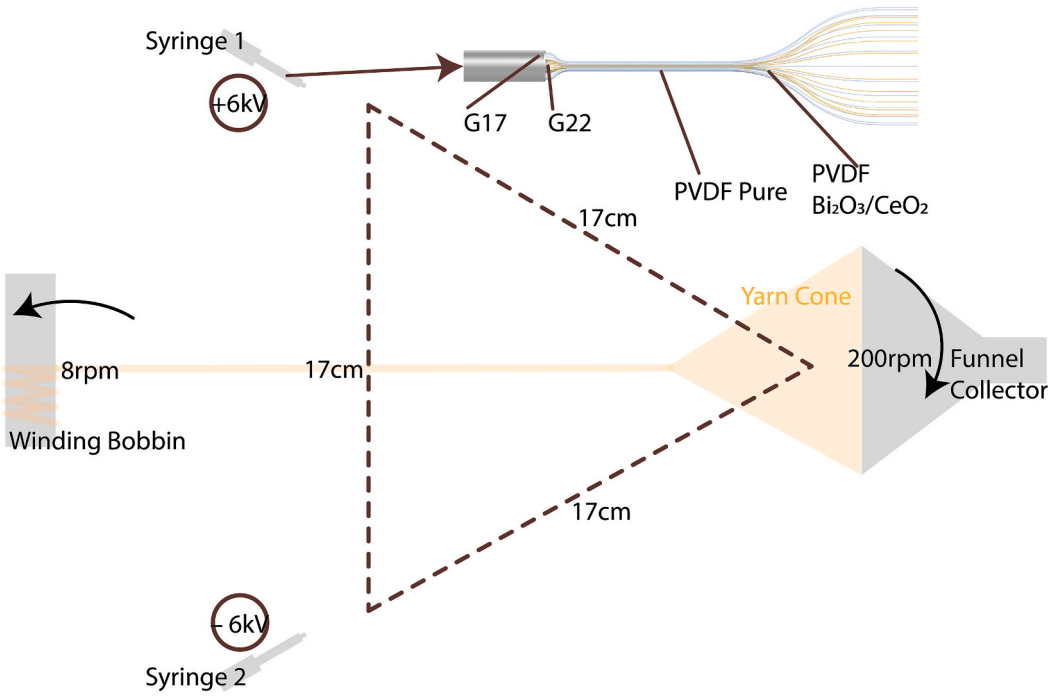


Fig. 2. Schematic of the coaxial dual-needle electrospinning setup with opposed-polarity configuration (±6 kV) and funnel collector for continuous yarn formation.

Table 1
Summary of electrospun sample compositions and fabrication parameters.

Sample	Format	PVDF&BCRatio	BC wt%	Needle	Gauge
PVDF-F	Membrane	10:0	0	Regular	G22
PVDFBC-F	Membrane	5:5	50	Regular	G22
PVDF-Y	Yarn	10:0	0	Regular	G22
PVDFBC-Y55	Yarn	5:5	50	Coaxial	G22/G17
PVDFBC-Y37	Yarn	3:7	70	Coaxial	G22/G17
PVDFBC-Y19	Yarn	1:9	90	Coaxial	G22/G17

410 Versa (80 kV, 10 W). Characterization was conducted for two purposes. The comparison with a commercial 0.2 mm lead film about attenuation-relevant conditions. Second, a 3D scan of the yarn and fibre structure, which is inside the yarn.

2.3.3. ATR-FTIR spectroscopy

ATR-FTIR spectra were conducted using a PerkinElmer Spectrum 3 FTIR spectrometer equipped with a Specac Golden Gate ATR accessory over the range 500–2000 cm^{-1}. The electroactive β-phase fraction F(β) was calculated from the absorbance intensities at 840 cm^{-1} (β-phase) and 763 cm^{-1} (α-phase) using the equation proposed by Gregorio and Cestari[9]:

$$F(\beta) = \frac{A_\beta}{(K_\beta/K_\alpha)A_\alpha + A_\beta} \quad (1)$$

$K_\alpha = 6.1 \times 10^4$ cm^2/mol and K_β = 7.7×10^4 cm^2/mol are the absorption coefficients.

2.3.4. Differential scanning calorimetry (DSC)

Thermal analysis was performed using a PerkinElmer DSC 4000 differential scanning calorimeter. Samples were heated at 10 °C min^{-1} and the melting temperature (T_m) was determined from the peak of the endothermic melting transition. Crystallinity (χ_c) was calculated as:

$$\chi_c = \frac{\Delta H_m}{\Delta H_m^0 \cdot w_{PVDF}} \times 100\% \quad (2)$$

where ΔH_m is the measured enthalpy of fusion, $\Delta H^\circ_m = 104.7$ J/g is the theoretical enthalpy of 100% crystalline PVDF [9], and W_{PVDF} is the mass fraction of PVDF in the sample.

2.3.5. Thermogravimetric analysis (TGA)

Using a PerkinElmer TGA 4000 under nitrogen atmosphere from 100 °C to 725 °C at a heating rate of 10 °C min^{-1}. The thermal decomposition temperature is ($T_{5\%}$, defined as the temperature at 5% mass loss), and residual inorganic content at 600 °C for each sample.

2.3.6. Tensile testing

Tensile properties using an Instron 5544 universal testing machine (Instron, Norwood, MA, USA) in accordance with BS EN ISO 5079:2020, at a pre-tension rate of 50% gauge length min-1 and a test extension rate of 100% gauge length min-1. BS EN ISO 5079:2020 was selected as the appropriate standard for single-fibre and yarn tensile characterisation in accordance with the yarn-format samples produced in this study. Young's modulus was determined from the slope of the initial linear elastic region of the stress–strain curve, ultimate tensile strength (UTS) from the peak stress, and elongation at break from the strain at fracture. Ten specimens per composition were tested. Specimens showing progressive fibre pull-out without a definitive fracture event. This behavior was more prevalent at higher filler loadings, yielding valid counts of n = 9 (PVDF-Y), n = 10 PVDFBC-Y55), n = 7 (PVDFBC-Y37), and n = 5 (PVDFBC-Y19).

3. Results and discussion

3.1. Electrospinning morphology

3.1.1. Membrane characterisation (Pre-Yarn Validation)

Electrospinning membranes of PVDF-F and PVDFBC-F were characterised by SEM to validate spinnability and particle dispersion prior to yarn production (Fig. 3). PVDF-F membranes exhibited uniform, bead-free fibres with a well-defined open porous network (Fig. 3a), consistent with stable electrospinning from the DMSO/acetone solvent system. Upon incorporation of Bi_2O_3 and CeO_2 nanoparticles at 50 wt% loading (PVDFBC-F), the membranes displayed increased fibre diameter and

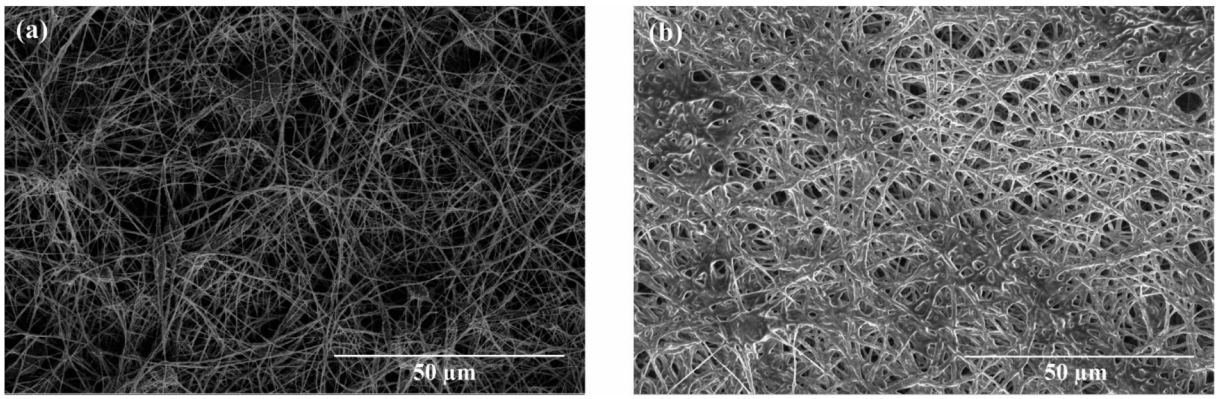


Fig. 3. SEM micrographs of electrospun membranes: (a) PVDF-F showing uniform, bead-free fibres with open porous network; (b) PVDFBC-F (50 wt% Bi_2O_3/CeO_2) showing increased fibre diameter and nanoparticle dispersion without large agglomerates.

partial inter-fibre fusion (Fig. 3b), indicative of nanoparticle-induced changes in solution viscosity and electrical conductivity[18]. This phenomenon of the fibre adhesion results from the static collection of the membranes: without the rotational drawing used in yarn formation, fibres accumulate at a fixed position and retain residual solvent, adhering at contact points. SEM confirmed adequate particle dispersion within the polymer matrix, with no large-scale agglomerates visible at the magnifications examined.

The successful formation of bead-free fibres at 50 wt% filler loading is notable because previous studies have reported bead defects in single-fluid electrospinning at substantially lower concentrations, typically below 30 wt%, when solvent ratios are not optimised [17]. The absence of beads at 50 wt% in the present work demonstrates that the DMSO/acetone binary solvent system provides sufficient solution stability to accommodate high inorganic particle concentrations whilst maintaining fibre uniformity. This membrane-stage validation confirmed the viability of the composite solution for subsequent yarn electrospinning, where additional process complexity (funnel collection, twist assembly) imposes further demands on solution stability.

3.1.2. *Yarn morphology*

At 100 × magnification, all four yarns exhibited a twisted fibre bundle morphology characteristic of funnel-collected electrospun yarn (Fig. 4a, c, e, g), consistent with the structures reported by researchers, Ali et al. and Li et al[1]; N. [19], The fibre twisting angle was generally moderate and visually comparable across compositions. The yarn indicates that the Bi_2O_3/CeO_2 filler did not disrupt the fundamental yarn formation mechanism of the funnel collector.

Fibre diameters were measured from high-magnification SEM micrographs (n = 30 per composition) using ImageJ. PVDF-Y exhibited the smallest mean fibre diameter (2.77 ± 0.37 μm), consistent with the absence of inorganic filler. Adding Bi_2O_3/CeO_2 nanoparticles increased fibre diameter, but the trend was non-monotonic: PVDFBC-Y55 (3.54 ± 0.62 μm) showed the largest increase, PVDFBC-Y37 (3.30 ± 0.40 μm) was slightly smaller, and PVDFBC-Y19 (3.77 ± 0.53 μm) increased again. Two competing effects account for this pattern. Higher filler loading raises solution viscosity and reduces the net charge density available for electrostatic stretching, both of which favour thicker fibres (D. [18]. However, from PVDFBC-Y55 to PVDFBC-Y37 the PVDF mass fraction in the core solution drops from 50% to 30%, reducing the volume of continuous polymer scaffold available to form each fibre. This reduction in polymer content offsets the viscosity-driven thickening, producing a net decrease in fibre diameter at 70 wt% loading. At 90 wt% (PVDFBC-Y19), the sheer volume of inorganic particles dominates: the particles physically expand the fibre cross-section despite the minimal polymer content (10 wt% PVDF), yielding the largest mean diameter in the series.

Despite filler loadings ranging from 0 to 90 wt%, the overall diameter increase remained within 19–36% relative to pure PVDF-Y. This narrow range suggests that the coaxial sheath configuration confines the diameter-expanding effect predominantly to the core channel, while the PVDF sheath maintains a consistent contribution to the overall fibre dimension.

At 5,000 × magnification, PVDF-Y fibres presented smooth, uniform surfaces with no visible particulate features (Fig. 4b), confirming that the neat PVDF solution produced a clean matrix free of contamination. Progressive changes in surface morphology were observed with increasing filler content. PVDFBC-Y55 displayed sparse bright-contrast

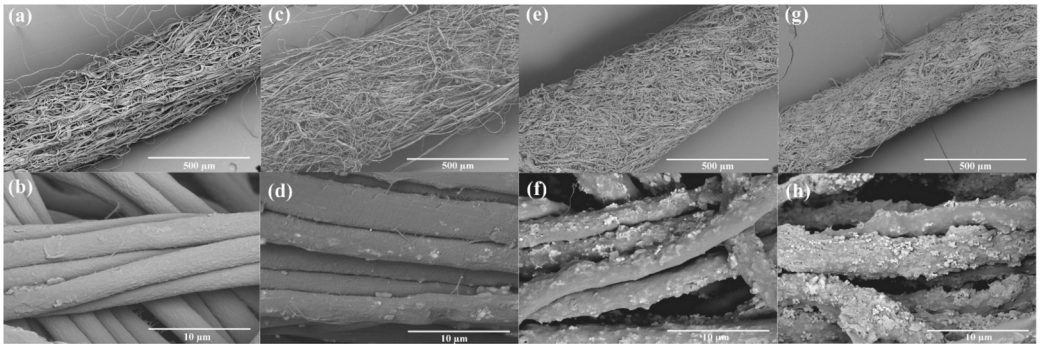


Fig. 4. SEM micrographs of electrospun yarn compositions (CBS detector, All mode). Upper row (a, c, e, g): low-magnification overview (scale bar = 500 μm) showing twisted fibre bundle morphology across all compositions. Lower row (b, d, f, h): high-magnification surface detail (scale bar = 10 μm) of (b) PVDF-Y, (d) PVDFBC-Y55, (f) PVDFBC-Y37, and (h) PVDFBC-Y19. PVDF-Y fibres exhibit smooth, uniform surfaces with no visible particulate features. With increasing Bi_2O_3/CeO_2 filler content, progressive accumulation of bright contrast features is observed, with sparse surface deposits in PVDFBC-Y55 (d), increased particle density in PVDFBC-Y37 (f), and dense surface protrusions with extensive agglomeration in PVDFBC-Y19 (h).

features on the fibre surface (Fig. 4d), indicating that only a small fraction of inorganic particles migrated to the fibre exterior. In PVDFBC-Y37, the density of surface-exposed particles increased markedly (Fig. 4f). PVDFBC-Y19, at the highest filler loading, revealed a dense distribution of bright protrusions with localised agglomeration across the fibre surface (Fig. 4h).

This progressive surface exposure of nanoparticles carries two implications. First, it provides independent morphological evidence that the core–sheath boundary becomes increasingly diffuse at higher filler loadings: a finding later corroborated by XCT reconstruction (Section 3.3). As the core solution carries more inorganic content relative to polymer, the coaxial interface loses definition, allowing particles to penetrate the PVDF sheath during jet formation. Second, this surface exposure does not compromise the intended shielding function, because X-ray attenuation is governed by the total attenuating mass within the beam path across the full fibre cross-section, not by whether the particles reside in the core or at the surface.

3.2. X-ray shielding performance

X-ray shielding performance was evaluated by XCT radiography using a ZEISS Xradia 410 Versa (80 kV, 10 W). Because the samples differed in geometry: the lead reference and PVDFBC-F were flat films, whereas PVDFBC-Y55, Y37, and Y19 were cylindrical yarns, each specimen was scanned individually under identical beam parameters to ensure comparability of normalised transmitted intensity values. Each attenuation value in Table 2 represents a single XCT acquisition without replicate-derived uncertainty bounds; this limitation is acknowledged and discussed below.

The transmission coefficient T was defined as:

$$T = I_{min}/I_0 \tag{3}$$

where I_m in is the minimum recorded intensity at the sample centre and I_0 is the mean background intensity measured on both sides of the sample.

The effective linear attenuation coefficient μ_eff was estimated via the Beer-Lambert law:

$$\mu_{eff} = -\ln(T)/d \tag{4}$$

where d is the yarn diameter measured from SEM micrographs (Table 2).

3.2.1. Membrane & lead reference (Fig. 5a)

The commercial 0.2 mm lead film and the PVDFBC-F composite membrane performed nearly identically at 80 kV, recording transmission coefficients of T = 0.522 (47.8% attenuation) and T = 0.513 (48.7% attenuation), respectively (Fig. 5a; Table 2). In terms of μ_{eff}, the composite membrane (0.409 mm^{-1}) exceeded the lead film (0.330 mm^{-1}) by approximately 24%, reflecting the difference in sample thickness: the lead film (1,972 μm) was thicker than the electrospun membrane (1,633 μm), so the composite achieved equivalent gross attenuation through a shorter path length. This result establishes that the PVDF/Bi_2O_3/CeO_2 material system attenuates X-rays at least as effectively as lead on a per-unit-thickness basis at 80 kV, validating the material composition before proceeding to the more challenging yarn format.

Table 2
X-ray attenuation parameters derived from XCT intensity profiles. I0: mean background intensity; I_{min}: minimum transmitted intensity at sample centre; T: normalised transmission coefficient; d: estimated sample thickness; μ_{eff}: effective linear attenuation coefficient (Beer–Lambert law).

Sample	I_0	I_{min}	T	Attenuation (%)	d (μm)	μ_{eff} (mm^{-1})
Lead film	411.3	214.6	0.522	47.8	1972	0.330
PVDFBC-F	452.1	231.9	0.513	48.7	1633	0.409
PVDFBC-Y55	2525.0	1241.0	0.491	50.9	1146	0.620
PVDFBC-Y37	1276.8	798.8	0.626	37.4	782	0.600
PVDFBC-Y19	1757.1	842.8	0.480	52.0	960	0.766

3.2.2. Yarn shielding performance and the role of yarn diameter (Fig. 5b, c)

The normalised intensity profiles in Fig. 5c illustrate the X-ray transmission through each sample. The profile width reflects the sample thickness along the beam path, while the depth of the trough indicates the degree of attenuation: a deeper and narrower trough corresponds to more efficient shielding per unit thickness. The minimum of each curve corresponds to the T value reported in the Table 2.

Among the three yarns, raw transmission T did not follow a monotonic trend with filler loading: PVDFBC-Y55 (T = 0.491, 50.9%), PVDFBC-Y37 (T = 0.626, 37.4%), and PVDFBC-Y19 (T = 0.480, 52.0%). This apparent inconsistency is resolved by accounting for yarn diameter. PVDFBC-Y37 has the smallest diameter in the series (782 μm versus 1,146 μm for PVDFBC-Y55 and 960 μm for PVDFBC-Y19), which shortens the absolute attenuation path length and directly reduces gross attenuation regardless of composition. The origin of this diameter variation is compositional: as the PVDF:BC mass ratio shifts from 5:5 to 3:7, the polymer fraction available to form the continuous fibre scaffold decreases, producing thinner individual fibres that twist into a smaller yarn cross-section. The electrospinning and winding parameters were held constant across all compositions, so the diameter reduction is attributable entirely to the reduced polymer content.

PVDFBC-Y19, despite having the lowest PVDF content (10 wt%), exhibited a larger yarn diameter (960 μm) than Y37. At 90 wt% filler loading, the inorganic particles occupy the majority of the fibre volume and produce a rough, loosely packed fibre morphology (as observed by SEM, Section 3.1.2). These fibres resist tight consolidation during twist assembly, forming a yarn cross-section that is larger but less dense than Y37.

When thickness-normalised through μ_eff, the trend becomes monotonic and physically consistent: μ_{eff} increases from 0.620 mm^{-1} (PVDFBC-Y55) to 0.600 mm^{-1} (PVDFBC-Y37) to 0.766 mm^{-1} (PVDFBC-Y19). The near-identical μ_{eff} of PVDFBC-Y55 and PVDFBC-Y37 (differing by only 3%) suggests that between 50 and 70 wt% loading, additional Bi_2O_3/CeO_2 content is partially offset by increased particle agglomeration, which introduces local compositional inhomogeneity and limits further gains in volumetric shielding efficiency. Beyond this plateau, the substantial increase in μ_eff for Y19 (27% higher than PVDFBC-Y37) indicates that at 90 wt% the sheer density of high-Z material overcomes the agglomeration penalty, consistent with the photoelectric absorption cross-sections of Bi (Z = 83) and Ce (Z = 58) dominating X-ray interaction at 80 kV.

Fig. 5c also reveals that the PVDFBC-F membrane (dashed purple curve) and the commercial lead film (solid black curve) show similar trough depths but different profile widths, consistent with comparable gross attenuation being achieved through different path lengths. Among the yarns, PVDFBC-Y19 (dotted orange curve) exhibits the deepest trough relative to its profile width, visually confirming its highest μ_{eff}. PVDFBC-Y37 (dash-dot pink curve) shows the narrowest profile, reflecting its smallest diameter with a correspondingly shallower trough, while PVDFBC-Y55 (solid yellow curve) presents the widest yarn profile with intermediate trough depth.

These results demonstrate that the PVDF/Bi_2O_3/CeO_2 composite retains effective X-ray shielding capacity when transferred from a flat membrane to a continuous yarn format. The intrinsic shielding efficiency (μ_{eff}) improves with increasing filler loading, while the gross attenuation *(T)* for any given yarn can be independently tuned by adjusting yarn diameter through electrospinning and winding parameters. This decoupling of composition-dependent shielding efficiency

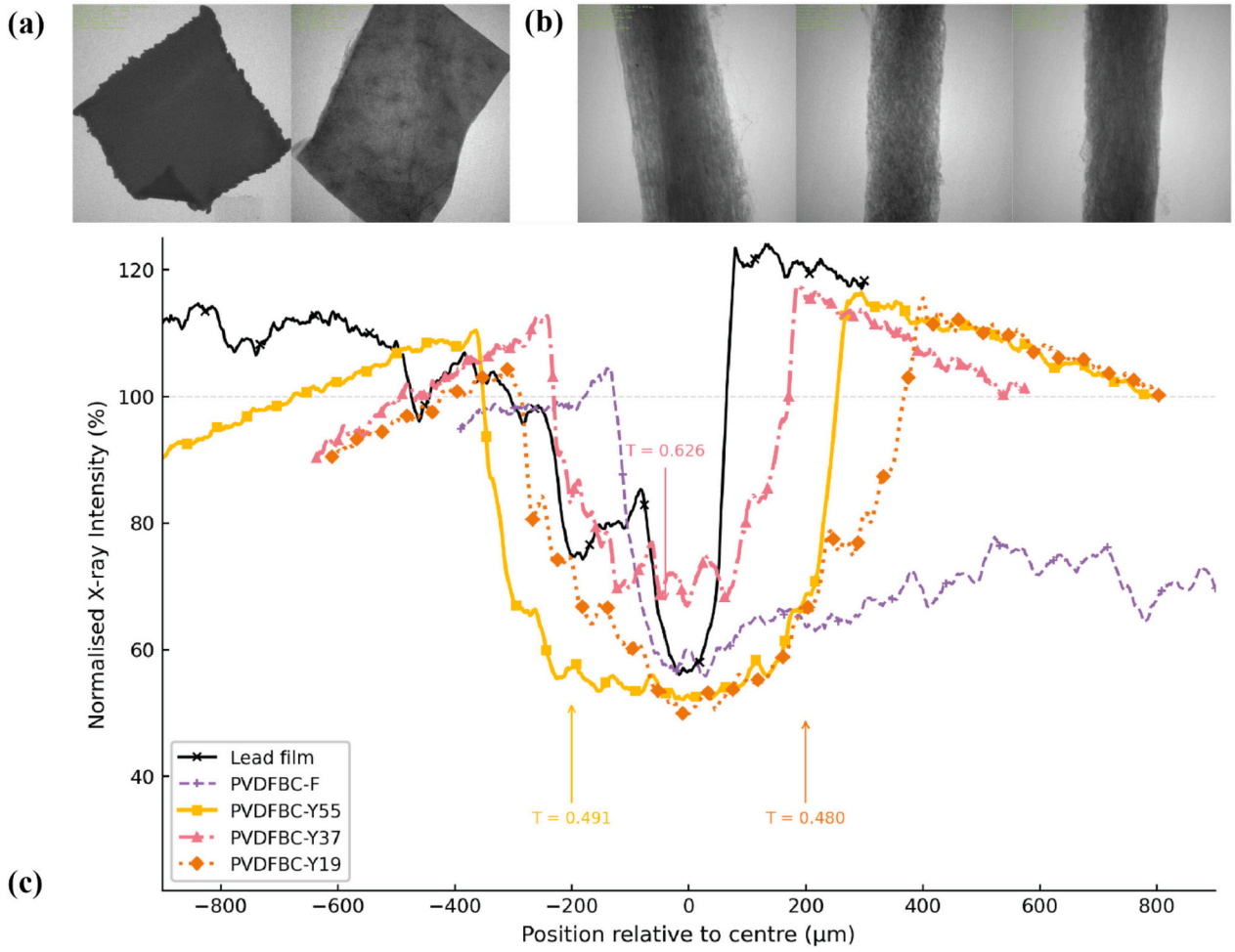


Fig. 5. XCT radiography of PVDFBC composite samples (ZEISS Xradia 410 Versa, 80 kV, 10 W). (a) XCT images of commercial lead film (left) and PVDFBC-F composite film (right). (b) XCT images of composite yarns PVDFBC-Y55 (left), PVDFBC-Y37 (centre), and PVDFBC-Y19 (right). (c) Normalised X-ray transmission intensity profiles of all samples, aligned at minimum intensity point and normalised to local background (I_0).

from geometry-dependent path length is an important design feature: it allows future optimisation of yarn diameter for textile processability without sacrificing per-unit-thickness attenuation. Although the loosely consolidated yarn format has an inherently lower packing density than the highly consolidated membrane (PVDFBC-F, $\mu_{eff} = 0.409$ mm^{-1}), all three yarns show higher intrinsic attenuation efficiency than the membrane (Table 2). This confirms that the loose packing of the yarn geometry does not penalize shielding effectiveness on a per-thickness basis.

The Bi_2O_3/CeO_2 combination employed here follows the synergistic K-edge shielding strategy demonstrated by Li et al. [20] in Bi/Ce-impregnated natural leather. Alsaab and Zeghib achieved comparable attenuation with Bi_2O_3-loaded PMMA nanocomposite films [2]. The present results extend this material strategy to a continuous electrospun yarn of PVDF, a format not previously reported, confirming that the synergistic shielding effect of the Bi/Ce pair is preserved through the electrospinning and yarn formation process.

3.3. Core–sheath via XCT reconstruction

PVDFBC-Y37 was selected as the representative sample for three-dimensional XCT scanning because it occupies an intermediate filler loading (70 wt%) at which SEM had already revealed moderate surface particle exposure (Section 3.1.2). If radial compositional segregation holds at this intermediate loading, a similar or more pronounced core-sheath structure can reasonably be inferred for PVDFBC-Y55 (lower loading, less surface exposure) and Y19 (higher loading, extensive surface exposure confirmed by SEM).

Three-dimensional reconstruction was performed in 3D Slicer v5.1.0 (Fedorov et al., 2012) using volumetric data acquired from a ZEISS Xradia 410 Versa micro-CT system (80 kV, 10 W). The high-attenuation phase (rendered in gold, corresponding to the BC-dominated fibre population) and the low-attenuation phase (rendered in cyan, corresponding to the PVDF-dominated fibre population) were separated by manual greyscale thresholding.

Fig. 6a presents the overview reconstruction of the PVDFBC-Y37 yarn segment, showing the spatial distribution of both phases along the yarn axis. The gold (BC-rich) and cyan (PVDF-rich) regions are visibly interleaved, confirming that the twist-based assembly mechanically intermingles the constituent fibre populations rather than producing a simple concentric bilayer.

At the yarn scale, side-view and semi-transparent renders (Fig. 6b-d) reveal that the BC-dominated fibres concentrate towards the yarn centre while PVDF-dominated fibres are preferentially located at the periphery. This radial compositional gradient arises from three co-existing fibre populations produced by the coaxial dual-needle system: coaxial core-sheath fibres (BC core with PVDF sheath), homogeneous BC-loaded fibres from the inner needle, and neat PVDF fibres from the outer needle. Twist assembly draws the denser, BC-rich fibres inward and positions the mechanically stronger PVDF fibres at the surface, a configuration that is functionally advantageous, as the exterior fibres bear the tensile and abrasion stresses encountered during subsequent textile processing

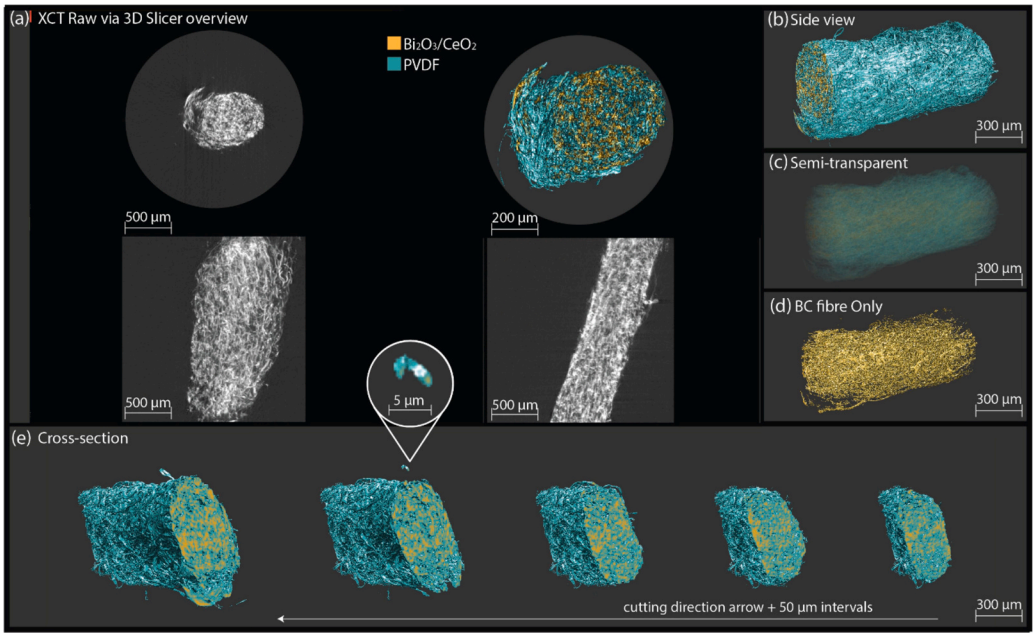


Fig. 6. XCT three-dimensional reconstruction of PVDFBC-Y37 sample via ZEISS Xradia 410 Versa, 80 kV, 10 W; reconstructed and rendered in 3D Slicer v5.1.0. The gold phase represents the Bi_2O_3 and CeO_2 filler region, corresponding to the higher X-ray attenuation area and the BC-dominated fibre population; the cyan phase represents the lower X-ray attenuation region, corresponding to the PVDF-dominated fibre population. (a) 3D Slicer overview XCT image (b) Side view of dual-phase BC core twisted and enclosed by PVDF outer layer. (c) Semi-transparent render showing BC and PVDF interio yarn twisted. (d) BC fibre population isolated after PVDF via transparency rendering. (e) cross-section showing BC concentrated toward the yarn centre this is confirming core–sheath morphology at both fibre and yarn level.

(warping, weaving, handling).

At the single-fibre scale, the cross-sectional slice (Fig. 6e) shows clear core-sheath morphology within individual fibres, with a BC-rich core enclosed by a PVDF-dominated sheath. This observation is consistent with three independent lines of evidence: XCT attenuation mapping shows relatively uniform attenuation across the yarn cross-section (indicating high-Z material is distributed throughout the interior); SEM reveals progressively increasing but still limited particle exposure on the fibre surface (Section 3.1.2); and the characteristic yellow colour of the as-spun yarn results from the dense BC core being visible through the translucent PVDF exterior.

The XCT reconstruction confirms that the yarn architecture achieves the intended design function: shielding capacity is maximised in the yarn interior, while a mechanically intact PVDF surface is maintained on

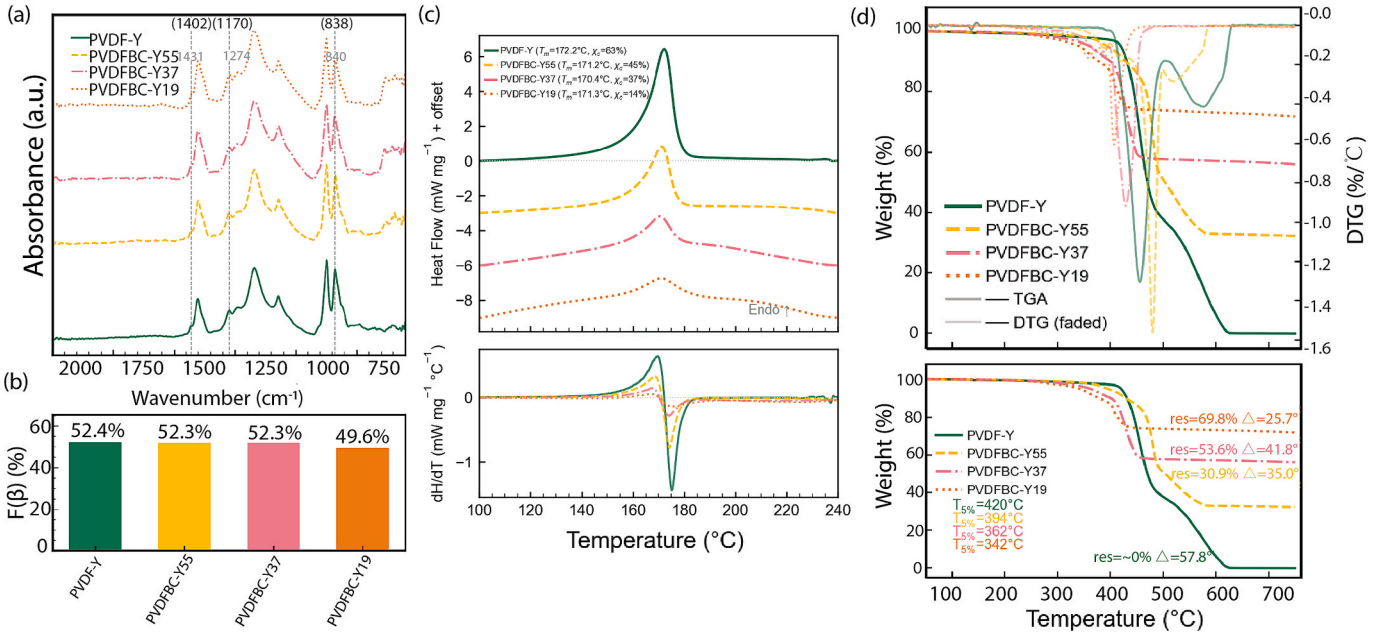


Fig. 7. Thermal and structural characterisation of PVDF-Y and PVDFBC composite yarns. (a) ATR-FTIR spectra (500–2000 cm^{-1}) with characteristic β-phase peak at 840 cm^{-1} and α-phase peak at 763 cm^{-1} annotated. (b) β-phase fraction F(β) calculated from FTIR absorbance ratios. (c) DSC thermograms (upper) and derivative heat flow dH/dT (lower) showing melting temperature Tm and degree of crystallinity χc for each composition. (d) TGA curves showing weight loss (upper, full range 100–725 °C) and expanded decomposition region (lower) with onset temperatures and residual inorganic content annotated.

the outside. The diffuse nature of the core–sheath interface, evident in both the SEM surface morphology (Section 3.1.2) and the XCT greyscale gradients, is attributed to the identical flow rates used for the core and sheath channels during electrospinning. Varying the core-to-sheath flow rate ratio is a straightforward route to sharpening this boundary in future work.

3.4. Thermal characterization

3.4.1. β-Phase Content (FTIR)

ATR-FTIR spectra of all four yarn compositions are presented in Fig. 7a. Across the full spectral range (500–2000 cm^{-1}), all samples exhibit similar overall absorption patterns, confirming that the PVDF matrix retains its characteristic chemical structure regardless of filler loading. The most prominent spectral features relevant to electroactive phase identification are the β-phase absorption at 840 cm^{-1} and the α-phase absorption at 763 cm^{-1}, both of which are clearly resolved in every composition. No additional peaks attributable to Bi_2O_3 or CeO_2 were observed in this spectral range, indicating that the inorganic fillers do not introduce new chemical bonding interactions with the PVDF matrix detectable by FTIR.

The electroactive β-phase fraction F(β) was calculated from the absorbance intensities at 840 and 763 cm^{-1} using Eq. (1) (Section 2.3.3). As shown in Fig. 7b, F(β) decreased only marginally from 52.4% (PVDF-Y) to 49.6% (PVDFBC-Y19), remaining highly stable across the full range of filler loadings. This stability demonstrates that the DMSO/acetone binary solvent system effectively promotes β-phase formation even at filler loadings as high as 90 wt%, with the resulting F(β) values fully comparable to those reported for DMF-electrospun PVDF fibres (45–60%) [12]. Replacing DMF with DMSO/acetone, therefore does not compromise electroactive performance while eliminating the reproductive toxicity associated with DMF. It is a key occupational health motivation for this solvent substitution [31].

The coaxial architecture is considered to contribute to this stability: the pure PVDF sheath crystallises under the same electrospinning-induced stretching and rapid solvent evaporation conditions that favour β-phase nucleation, so that even as the core becomes increasingly occupied by inorganic filler, the fibre surface, where ATR-FTIR preferentially samples: retains high β-phase content.

3.4.2. Crystallinity (DSC)

DSC thermograms (Fig. 7c, upper panel) show a single endothermic melting peak for all compositions, with the derivative heat flow (Fig. 7c, lower panel) used to determine the melting temperature T_m. The degree of crystallinity χ_c was calculated from Eq. (2) (Section 2.3.4) and fell steadily with increasing filler loading: 62.6% (PVDF-Y), 44.6% (PVDFBC-Y55), 36.9% (PVDFBC-Y37), and 13.9% (PVDFBC-Y19). This progressive decline is attributed to the inorganic particles physically disrupting chain folding and lamellar organisation within the PVDF matrix, reducing the fraction of polymer that can crystallise as filler content increases.

The melting temperature T_m decreased slightly from PVDF-Y to the intermediate compositions, then partially recovered for PVDFBC-Y19. At 90 wt% loading, the high particle density physically confines PVDF molecular chains, restricting segmental mobility and effectively raising the energy required for melting. This confinement effect, combined with heterogeneous nucleation by the nanoparticle surfaces [3], accounts for the non-monotonic T_m trend.

The decoupling of crystallinity and β-phase fraction χ_c drops while F(β) holds steady is a key finding. It indicates that the crystalline PVDF that does form is consistently β-dominated, regardless of how much total crystallinity is lost to filler disruption. Whether this reduced but β-rich crystalline fraction generates a sufficient piezoelectric response for sensing applications remains to be confirmed by direct d_{31} measurement in future work [9].

3.4.3. Thermal stability (TGA)

TGA was conducted under nitrogen from 100 to 725 °C to assess whether the yarns can withstand steam autoclave sterilisation (134 °C), a standard requirement for reusable clinical textiles. As shown in Fig. 7d, all compositions began thermal decomposition between 420 and 450 °C, which is well above the 134 °C autoclave threshold, confirming that repeated sterilisation cycles will not thermally degrade these yarns.

The onset of decomposition ($T_{5\%}$) decreased with filler loading, from 420 °C for PVDF-Y to 342 °C for PVDFBC-Y19. This reduction reflects the lower proportion of thermally stable PVDF matrix relative to the total sample mass; however, even the lowest $T_{5\%}$ value (342 °C) exceeds clinical sterilisation requirements by more than 200 °C.

Residual mass at 600 °C increased progressively from PVDFBC-Y55 to Y37 to Y19, consistent with the nominal inorganic filler proportions in each composition. This independent gravimetric confirmation that the intended Bi_2O_3/CeO_2 content was successfully incorporated into the yarns complements the compositional evidence from SEM (Section 3.1.2) and XCT (Section 3.3). Under nitrogen, the residue comprises both the inorganic filler and carbonaceous char from PVDF pyrolysis.

4. Summary

In conclusion, FTIR demonstrates that the β-phase is retained at high filler loadings. DSC shows the crystallinity decreases with increasing bismuth oxide and cerium oxide, with remaining predominantly β-phase. TGA reveals the yarns are thermally stable at intended clinical application in sterilization temperatures, whilst independently verifying the filler content across compositions. Taken together, these three results demonstrate that the coaxial architecture preserves the functional material properties of PVDF even under the extreme condition of 90 wt % filler loading.

4.1. Mechanical properties

Tensile properties of the four yarn compositions were tested on an Instron universal testing machine following BS EN ISO 5079:2020 (gauge length 20 mm, extension rate 100%/min, n = 10 per composition). The results are presented in Fig. 8.

UTS decreased from 9.83 MPa for PVDF-Y to 6.05 MPa for PVDFBC-Y55. More additive material means more disruption to the PVDF, and a weaker yarn. Elongation at break followed the same trend, from 311.8% to 83.9%. PVDFBC-Y37, however, showed higher elongation (282.4%) than PVDFBC-Y55 (164.8%). At this middle point, interfacial adhesion between core and sheath is insufficient to hold fibres together, so they pull out one by one rather than fracturing as a bundle. PVDFBC-Y19 is dry and rough from the previous SEM image and the single-fibre pull-out with extensive fibre drawing. The failure mode was not consistent with the observation on the tensile test reality.

Young's modulus appeared in a V-shape in four samples. PVDF-Y (33.0 MPa) showed the lowest modulus among the four compositions. PVDFBC-Y55 (61.3 MPa) was higher, indicating that at 50 wt% filler loading, particle reinforcement more than compensates for the loss in crystallinity. However the 3:7 ratio, Young's modulus dropped to 25.9 MPa, as matrix disruption overtook the reinforcing effect. Y37 is the softest of the four samples. Y19 climbed back to 102.5 MPa, because at 90 wt% filler the densely packed particles themselves dictate stiffness. The yarn is harder but breaks more readily. This V-shaped trend reflects that polymer and additive materials balance each other, which defines a limitation for additive materials loading compatible with textile processing where yarn flexibility is essential.

Despite the decline in mechanical performance with filler loading, the UTS of all yarns falls slightly above the 1–10 MPa range reported by Mouthuy's research for continuous electrospun filaments [25]. The continuous neat PVDF sheath in the coaxial electrospinning architecture mechanically protects the heavily filled core. The dependence of yarn strength on processing and drawing conditions observed here is

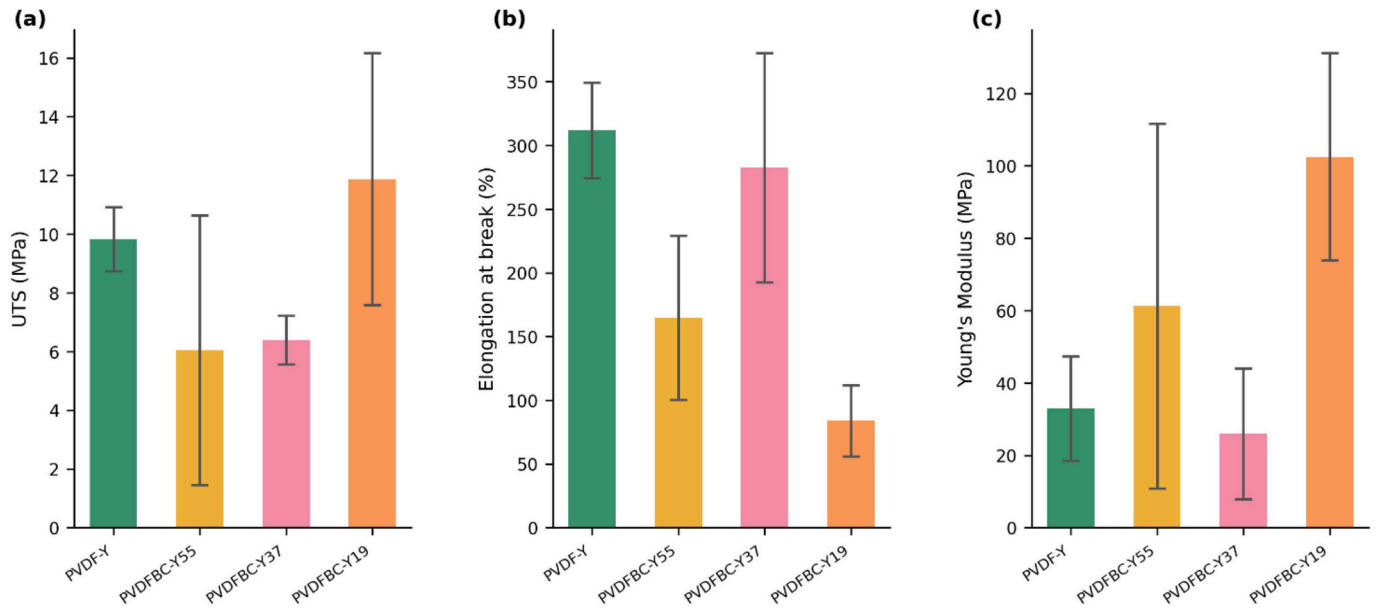


Fig. 8. Tensile properties of PVDF-Y and PVDFBC composite yarns: (a) ultimate tensile strength (UTS), (b) elongation at break, and (c) Young's modulus. Results presented as mean ± standard deviation. Valid specimen counts after exclusion of indeterminate records: PVDF-Y (n = 9) and PVDFBC-Y55 (n = 10), PVDFBC-Y37 (n = 7), PVDFBC-Y19 (n = 5).

consistent with recent reports on electrospun PVDF nanofibre yarns [10]. The UTS and elongation values indicate that all four compositions can withstand typical warp tensions encountered during weaving. PVDFBC-Y55, retaining mechanical properties closest to unfilled PVDF-Y whilst incorporating 50 wt% shielding filler, may represent the optimal balance for initial weaving trials.

4.2. Comparison with existing studies

Several lead-free shielding composites have been reported, including Bi/Ce-impregnated natural leather[20], PMMA/Bi_2O_3 bulk films[2], and electrospinning PVA/Bi_2O_3 nanofibre membranes[13]. These studies have demonstrated effective X-ray attenuation across various energy ranges. However, all of them remain in flat formats-films, membranes, or coated substrates, which cannot be woven, knitted, or sewn into garments. Highly filled polymer fibres have recently been produced for multilayer radiation-protective fabrics [21]. However, these rely on melt-based fibre filling rather than a continuous electrospun core–sheath architecture.

This study produces the first continuous radiation-shielding nanofibre yarn via electrospinning. Unlike flat formats, a yarn is directly compatible with established textile manufacturing processes, providing a route from shielding material to a wearable protective garment.

The coaxial coresheath addresses two limitations in previous work, Jamil et al. [13] reported spinning instability at 40 wt% Bi_2O_3 in a single-fluid PVA system. The coaxial system in this study reaches 90 wt% in the core and is wrapped with a pure PVDF sheath. The sheath serves a mechanical function, which maintains yarn integrity under tensile and abrasion loading, keeping UTS above the 1–10 MPa range reported for continuous electrospun filaments[25]. Without the sheath, fibres at this loading would be brittle and non-processable.

Table 3 compares the mechanical properties across three fibre formats relevant to this work. A yarn is a twisted bundle of nanofibres with inter-fibre porosity; a filament is an aligned bundle of nanofibres with inter-fibre porosity but without twist; and a single fibre is a dense monofilament produced by melt extrusion without internal porosity. The progressively lower UTS from single fibre to filament to yarn reflects increasing void fraction rather than differences in intrinsic material strength.

The present electrospun yarns (UTS = 6.05–11.87 MPa) fall within, even slightly above the 1–10 MPa range reported by Mouthuy et al. [25] for continuous electrospun PCL filaments. Although containing up to 90 wt% additive materials (Bi_2O_3 and CeO_2), the PVDFBC yarns do not fall below this baseline, demonstrating that high additive material loading does not compromise the fundamental mechanical viability of the electrospinning yarn format. Fan et al. achieved a raw pure PVDF

Table 3
Comparison of mechanical properties with literature-reported PVDF fibres and films.

Sample	Formation	Method	UTS (MPa)	Elongation (%)	Young's Modulus (MPa)	Diameter (μm)
PVDF-Y	Yarn	Electrospinning Funnel/Conjugate	9.83 ± 1.08	311.8 ± 37.2	33.0 ± 14.4	472.1
PVDFBC-Y55	Yarn	Electrospinning Funnel/Conjugate Coaxial	6.05 ± 4.60	164.8 ± 64.6	61.3 ± 50.4	566.0
PVDFBC-Y37	Yarn	Electrospinning Funnel/Conjugate Coaxial	6.40 ± 0.83	282.4 ± 89.9	25.9 ± 18.1	546.1
PVDFBC-Y19	Yarn	Electrospinning Funnel/Conjugate Coaxial	11.87 ± 4.29	83.9 ± 28.1	102.5 ± 28.6	409.2
PCL from [25]	Filament	Electrospinning Wire guide	1–10	N/A	N/A	N/A
PVDF from [8]	Yarn (raw)	Electrospinning Funnel/Conjugate	30.2	N/A	N/A	109
PVDF from [8]	Yarn (hot stretching)	Electrospinning Funnel/Conjugate	313.3	N/A	N/A	54
PVDF from [23]	Single Fibre	Melt Spinning	95	137	650	160

nanoyarn UTS of 30.2 MPa using a similar electrospinning set-up. The difference relative to the present PVDF-Y is attributed to the substantially smaller yarn diameter (109 μm vs 472 μm). Because it reduces the inter-fibre void fraction and increases the apparent UTS.

The yarn diameter in the present work is attributed to slower in funnel speed and winding rate, which affect fibre packing density and will be systematically investigated in future work. In Fan et al.'s further research, they demonstrated that hot stretching at 140 °C raises PVDF nanoyarn strength to 313.3 MPa, which indicates that post-drawing is a viable route to further mechanical enhancement of the present composite yarns in future work. Mokhtari et al. 's melt-spun PVDF single fibres exhibit a UTS of 35 MPa, indicating that the order-of-magnitude UTS difference between formats is structural rather than compositional. In conclusion, the current study's coaxial architecture preserves sufficient mechanical integrity for textile processing by adding the shielding materials via electrospinning process.

This work also replaces DMF, the conventional PVDF electrospinning solvent, with DMSO/acetone, removing reproductive toxicity without loss of β-phase content (49.6–52.4%). This occupational health concern has received limited attention in the literature on shielding.

5. Conclusion

Interventional radiology staff accumulates scattered radiation over entire careers. Conventional lead garments mitigate this exposure only at considerable physical cost. Aprons and thyroid collars weigh 2–7 kg. They restrict movement and cause musculoskeletal injury. Staff frequently leave them unworn as a result. This work set out to produce a lead-free, textile-processable shielding yarn that could eventually replace these garments. We selected DMSO over the more common DMF as the electrospinning solvent. DMF carries reproductive toxicity risks for laboratory and manufacturing personnel. DMSO proved compatible with stable fibre formation without measurable loss in fibre quality.

We electrospun three ratios of yarn through a coaxial dual-needle arrangement via electrospinning funnel collector. PVDFBC-Y55, PVDFBC-Y37 and PVDFBC-Y19 correspond to BC ratio loadings of 50, 70 and 90 wt% in the yarn. At 90 wt% loading, yarn formation would not have been possible without the coaxial sheath. The sheath PVDF supports the heavily filled core together during twisting. This structural role of the sheath strengthens simple mechanical reinforcement.

XCT reconstruction of the representative sample, PVDFBC-Y37, showed a radial compositional gradient. These are coaxial core-sheath fibres, homogeneously BC-loaded PVDF fibres, and neat PVDF fibres. The gradient is consistent with three independent observations. XCT attenuation maps are comparatively uniform across the cross-section. SEM confirms a high Z element fewer on the yarn surface.

X-ray Shielding measurements on the electrospinning PVDFBC-F membrane revealed a transmission factor of 0.513 at 80 kV. A commercial 0.2 mm lead film measured 0.522 under 80 kV X-ray power. The electrospinning membrane with the high Z element ratio effectively matches lead-equivalent attenuation. The 90 wt% variant recorded the highest value. β-phase fraction remained between 49.6% and 52.4% for every variant. Neither the coaxial spinning geometry nor the high filler loading disrupts the electroactive crystalline phase.

The current limitation centres on the core–sheath definition. XCT confirms a dual core-sheath architecture at both fibre and yarn level, but the boundary remains indistinct. Core and sheath solutions ran at identical flow rates during electrospinning, which likely accounts for the diffuse interface. A next step is to vary the core-to-sheath flow rate ratio across several conditions and examine whether a sharper macroscopic boundary results.

Characterisation of the PVDF β-phase so far relies on FTIR spectral analysis alone. Piezoelectric signal testing has not yet been carried out. Incorporating direct piezoelectric measurements is a priority for the next phase. Which biological signals the yarn can realistically detect also requires more detailed investigation.

CRediT authorship contribution statement

Siyuan Zhao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Lindsey Waterton Taylor:** Writing – review & editing, Supervision. **Vien Cheung:** Writing – review & editing, Supervision. **Yu Shi:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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